

Supplementary information: Shared burst dynamics and limits of neuron-specific identifiability in sparse-positive zebrafish fluorescence imaging

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Abstract

This supplement defines the complete identity, trace, control, statistical, and reproducibility contracts for the protected NeuRev confirmatory analysis. It also specifies the trace-atlas layout and the evidence required to promote functional, tensor, and detection results.

Working manuscript skeleton. Daggered values and figures are provisional outputs from the pre-repair repository. Author, ethics, funding, data-release, and final adjudication fields require review. The release build must disable provisional mode and contain no red placeholders.

S1. Extended methods and validation contract

S1.1. Stage artifact contract

Every analysis stage writes the following common files:

- `STAGE_SUMMARY.md`: scientific question, inputs, outputs, main findings, and limitations;
- `METRICS.json`: typed metrics with units, direction, uncertainty, and evidence status;
- `VALIDATION_REPORT.md`: test results, validity checks, warnings, and stop/downgrade decisions;

- `FIGURE_INDEX.md`: figure purpose, source data, generation command, and interpretation;
- `REVIEW_REQUIRED.md`: only unresolved decisions that require a person;
- `DECISIONS_REQUESTED.yaml`: structured options, recommended conservative default, affected outputs, and review paths;
- `source_manifest.json`: input hashes, code revision, configuration, environment, and output checksums.

A stage is not considered complete merely because its executable returned zero. It must satisfy its scientific validity checks and produce the common artifact set. Optional or conditionally skipped stages emit the same files with a machine-readable skip reason.

S1.2. Identity regression tests

The repaired identity layer must pass synthetic and repository-specific tests:

1. two records sharing a canonical identity but holding different site identifiers and coordinates produce two independent traces;
2. row ordering cannot change the selected geometry;
3. canonical aggregation after extraction preserves the source-site membership list;
4. a merge cannot reduce the number of stored source traces;
5. original-site and canonical views reproduce the same occurrence count;
6. every exported metric points to an observation identifier and site identifier;
7. the ROI 010/015 repository fixture fails under the historical dictionary implementation and passes under the repaired implementation.

S1.3. Trace-completeness checks

For each occurrence and mandatory representation, the trace inventory records frame range, spatial mask, units, transform, finite fraction, saturation fraction, baseline support, event support, and checksum. Mandatory traces are Raw, annulus, outer ring, residual, signed difference, and frozen ICA. A representation may be unavailable only when the stage declares a scientifically interpretable reason. Missing traces may not be silently omitted from denominators.

S1.4. Temporal-null construction

Admissible shifted windows match the labeled window duration and field context while excluding the union of all labeled windows plus a protected temporal buffer. Circular shifts are used only if wraparound has a meaningful acquisition interpretation; otherwise shifts are clipped to valid contiguous regions. Null samples are blocked by burst and site. The code records the exact shifted windows so that every empirical probability can be reconstructed.

For a statistic T whose larger value supports the alternative, the Monte Carlo upper-tail probability is

$$\hat{p} = \frac{1 + \sum_{r=1}^R \mathbb{I}(T_r^{\text{null}} \geq T^{\text{obs}})}{R + 1}. \quad (\text{S1})$$

The number of unique admissible shifts is reported. The manuscript does not imply more resolution than the null sample count supports.

S1.5. Spatial-control matching

Translated pseudo-ROIs are sampled within the same acquisition field and are matched on mask area, baseline intensity bin, boundary distance, and local saturation. A control is rejected if it overlaps a protected labeled region, leaves the image, or lacks the full annulus and outer ring. Each target receives the same requested number of controls unless geometric constraints make this impossible; control deficits are reported by site.

S1.6. Bootstrap hierarchy

The primary two-way bootstrap samples sites and bursts with replacement and then includes all available occurrence-level measurements within the selected site–burst cells. With only four bursts, intervals are supplemented by leave-one-burst-out estimates and all individual burst effects. A site-only bootstrap is used for metrics whose temporal null already conditions on burst. Pixel or frame bootstrap intervals are not used for neuron-level claims.

S1.7. Functional-analysis entry gates

Functional or tensor decomposition is executed only if all of the following hold:

1. at least 95% of included occurrences have complete mandatory event windows;
2. timing conventions are frozen for the primary view;
3. no single site or burst contributes more than a prespecified share of the squared norm after robust scaling;
4. basis rank or tensor rank is selected without the held-out burst;
5. component sign and permutation ambiguity are resolved using training data only;
6. held-out reconstruction improves over a shared-template baseline by the preregistered amount;
7. components are stable across at least three leave-one-burst-out fits.

Failure of an entry gate produces a descriptive shared-template analysis rather than forcing a high-capacity decomposition.

S1.8. Representation utility and fidelity

For each frozen representation, utility and fidelity are distinct outcome families. Utility includes known-positive recall at fixed budget, proposal-union coverage, nearest-candidate rank, and burden. Fidelity includes native-amplitude correlation, peak attenuation, area attenuation, peak-time shift, and waveform correlation. A representation cannot be promoted as a denoiser solely because it improves an onset score, nor as a detector solely because it preserves Raw morphology.

S1.9. Evidence levels

Results are assigned one of four evidence levels:

C0—descriptive: generated correctly but not tested against a protected null or held-out unit;

- C1—controlled:** compared with matched temporal or spatial controls within the canonical recording;
- C2—internally confirmed:** frozen before protected leave-one-burst-out or bounded-field evaluation;
- C3—externally confirmed:** reproduced on an independent recording acquired and evaluated under the frozen contract.

The current paper is expected to contain C1 and C2 results. C3 is a future promotion criterion, not an implicit claim.

S1.10. Autonomous execution and human feedback

The program continues through all unaffected stages. Identity, timing, annotation, authorship, ethics, funding, and release questions are collected into one end-of-run decision file. Each item includes a conservative default, the scientific consequence of each option, direct paths to review figures, stages that are blocked, and stages that remain valid. The default conservative behavior preserves original sites and original timing and labels unresolved canonical identity as unresolved.

S2. Extended current evidence and planned sensitivity analyses

S2.1. Current evidence inventory

The table below is an execution-oriented view of the current results and their required replacement artifacts. It is not a final statistical table.

Table S1: Analysis stages, primary estimands, and stage gates.

Stage	Workstream	Primary output	Continue/downgrade rule
00	Preflight and source inventory	Data, label, code, and output hashes; schema audit	Fatal only for corrupted or uninterpretable canonical inputs
01	Identity and geometry repair	Immutable site key; canonical mapping; regression tests	Must prove no spatial-site overwrite
02	Label and timing contract	Original/adjudicated timing views; review panels	Continue with original timing while decisions remain pending

Stage	Workstream	Primary output	Continue/downgrade rule
03	Raw-to-pipeline trace atlas	Complete trace/metric inventory for every occurrence	Missing mandatory trace or untracked unit blocks dependent analysis
04	Acquisition QC	Noise, saturation, drift, motion, field and optional-video reports	Optional videos may be absent; canonical QC is mandatory
05	Scalar observability	Variance components, ICCs, rank stability, burst sensitivity	Singular models downgrade to fixed/rank-based summaries
06	Spatial specificity	Radial profiles, matched controls, equivalence tests	Claims are downgraded if control matching or margins fail
07	Functional and tensor analysis	Shared, site, burst and residual temporal components	Run only after completeness and held-out stability gates
08	Measurement phenotype	Recoverability and failure-mode models	Clusters remain exploratory; continuous effects are primary
09	Bounded annotation	Blinded packet and local precision-recall	Nonblocking until completed reviewer labels return
10	Frozen representation panel	Utility, fidelity, burden and failure decomposition	No feature expansion; retain negative/equivalence results
11	Manuscript export	Macros, tables, figures and source maps	Every numerical claim must resolve to a validated source
12	Release validation	Reproducible package and aggregate decisions	Release fails if provisional or unresolved mandatory fields remain

S2.2. Burst-specific sensitivity

The protected output will report all primary scalar and trace metrics for each burst separately, with burst 2 highlighted as prespecified timing-uncertain. Required displays include:

- event-aligned Raw, annulus, residual, difference, and ICA population traces by burst;
- native-amplitude distributions and robust SNR by burst;
- original versus proposed timing overlays for burst 2;

- each primary effect with and without burst 2;
- leave-one-burst-out effect estimates and component stability;
- a statement of whether the substantive conclusion changes under any sensitivity view.

S2.3. Identity sensitivity

Until ROI 010/015 is adjudicated, the following result views are mandatory:

1. original-site analysis with all 27[†] coordinate-defined sites;
2. provisional canonical analysis with 26[†] identities, retaining separate source traces;
3. direct ROI 010-versus-ROI 015 trace and spatial comparison;
4. a hypothetical merged summary only if a defensible consolidated mask or aggregation rule is specified;
5. explicit identification of statistics that change under the merge view.

S2.4. Provisional signature figures

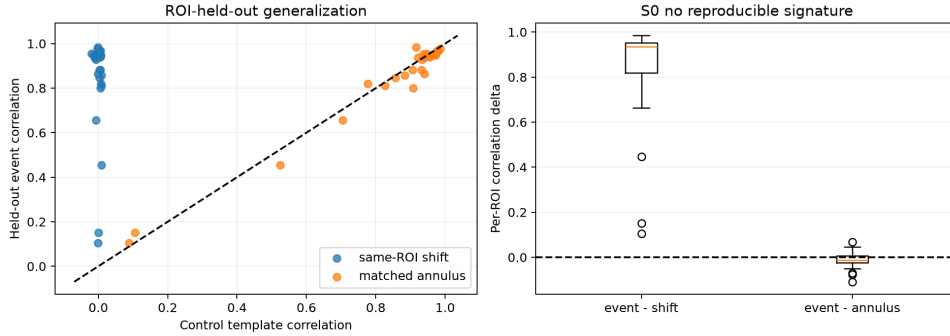


Figure S1: Existing Raw-signature generalization summary. This figure is retained as a visual reference only and will be replaced by the repaired held-out site analysis. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

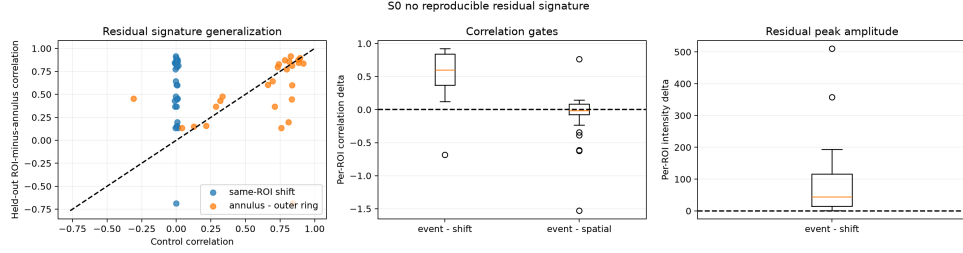


Figure S2: Existing residual-signature generalization summary. The replacement will separate original-site geometry from canonical identity and include matched translated spatial controls. **Provisional:** this artifact predates the identity/geometry repair and is included only to guide the confirmatory rerun.

S2.5. Negative and boundary results

The manuscript will preserve scientifically valid negative results. In particular, the final report will state when:

- a learned component is analytically equivalent to a simple operator;
- a repeatable event waveform is not spatially specific;
- a local residual has event amplitude but no stable shape;
- a functional or tensor model fails held-out stability;
- a representation improves known-positive recovery but materially degrades trace fidelity;
- an effect depends on burst 2, one field, one site, or one annotation choice;
- precision remains unidentified outside the bounded field.

These outcomes constrain future model design and should not be hidden by selective promotion.

S3. Trace-atlas and reviewer-packet specification

S3.1. Per-occurrence atlas page

Each labeled occurrence receives a stable page keyed by observation identifier. The page contains:

1. a full-frame context image with original site, canonical mapping, annulus, outer ring, nearest labeled sites, field boundary, and translated controls;
2. a high-resolution crop at baseline, proposed onset, Raw peak, residual peak, and event offset;
3. synchronized native-amplitude traces for Raw, annulus, outer ring, residual, difference, and ICA;
4. a separate robustly standardized trace panel for shape comparison;
5. onset, peak, offset, burst interval, saturation, and motion annotations;
6. a metric card with units and validity flags;
7. direct links to the source row, mask, trace array, configuration, and generation command.

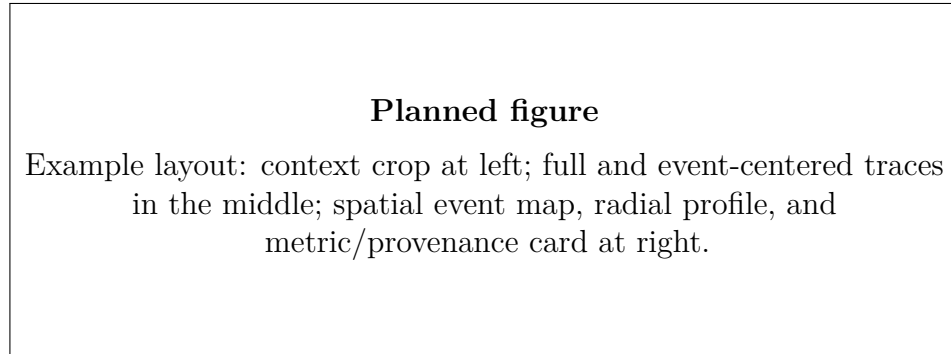


Figure S3: Planned per-occurrence trace-atlas page.

S3.2. Per-site summary

Each immutable site receives a neuron-by-burst summary showing all available occurrences without peak scaling. Required panels are:

- Raw and residual traces by burst;
- native amplitude and SNR by burst;
- spatial event maps and radial profiles;

- annulus coupling and specificity;
- candidate rank and recovery failure category;
- repeatability metrics and uncertainty;
- acquisition covariates and field membership;
- canonical-identity history and unresolved decisions.

S3.3. Reviewer packet

Review frames are generated independently of detector candidates to avoid circular adjudication. Reviewers receive original-intensity and contrast-normalized views, event movies, the original label, and a standardized response schema. They do not receive representation scores, candidate ranks, or model interpretations. The packet records completion state and supports re-review without changing the immutable source observations.

S3.4. Aggregate atlas index

The final atlas index supports filtering by burst, original site, proposed canonical identity, field, SNR quantile, morphology, saturation, annulus coupling, timing uncertainty, and recovery status. A machine-readable index mirrors every visual filter and prevents the HTML or PDF atlas from becoming the sole source of truth.